The angular momentum is mvd. Conservation gives:

$$mv_0(a + c) = mvd \tag{7}$$

The probe's energy at the hyperbola vertex equals that at infinity; thus

$$\frac{1}{2}mv_0^2 - \frac{Gm_E m}{a + c} = \frac{1}{2}mv^2 \tag{8}$$

From Fig. 20-2-2:

$$d = 2(a + c)\sin\alpha \tag{9}$$

A, B are transverse and conjugate semi-axes; C is the center-to-focus distance. Earth's center is a focus. The given data and analytic geometry give:

$$A = a + c \tag{10}$$

$$C = 2(a + c) \tag{11}$$

$$B^2 = C^2 - A^2 \tag{12}$$

and

$$\sin\alpha = \frac{B}{\sqrt{A^2 + B^2}} \tag{13}$$

These equations give:

$$\sin\alpha = \frac{\sqrt{3}}{2} \tag{14}$$

Solving (7), (8), (9), and (14):

$$v^2 = \frac{Gm_E}{a + c} \tag{15}$$

After ejection, the probe is in hyperbolic motion. Its energy is:

$$E_2 = \frac{1}{2}mv^2 = \frac{Gm_E m}{2(a + c)} \tag{16}$$

The energy given at launch is:

$$\Delta E = E_2 - E_1 \tag{17}$$

From (1), (6), and (16):

$$\Delta E = \frac{Gm_E m}{2a}\frac{2a + \sqrt{a^2 - b^2}}{a + \sqrt{a^2 - b^2}} \tag{18}$$

3.

At angular speed Ω, the emf induced in each spoke is

$$E = \frac{1}{2}l^2 B\Omega \tag{1}$$

If current I flows through R, each spoke carries

$$I' = \frac{1}{n}I \tag{2}$$

The magnetic force is:

$$F' = BlI' \tag{3}$$

The magnetic torque about the axle is

$$M_B' = \frac{1}{2}lF' \tag{4}$$

The total magnetic torque on the wheel is

$$M_B = nM_B' \tag{5}$$

When

$$M = M_B \tag{6}$$

the wheel rotates steadily: $\Omega = \Omega_0$, $I = I_0$. The above equations give

$$M = \frac{1}{2}Bl^2 I_0 \tag{7}$$

$$E = \frac{1}{2}Bl^2\Omega_0 \tag{8}$$

Ohm's law gives:

$$E = I_0 R \tag{9}$$

Solving (7), (8), and (9):

$$I_0 = \frac{2M}{l^2 B} \tag{10}$$